.things that Matter — Monday Meeting Brief

Date: Monday, 17 February 2025, 10:00 AM (Zoom) **Attendees:** Sobuhle Nyambali (CEO), Matome Mbowene (CTO)
Prepared by: CTO Office **Version:** 1.0

Executive Summary

This document covers 8 enterprise infrastructure items for discussion. Each section includes: what it is, why we need it, recommended tool/approach, estimated cost, and implementation timeline. Total estimated monthly cost for all items: **R850–R1,600/month** (can start at R0 with free tiers).

Meeting Agenda (30 min)

#	Item	Time	Priority
1	Quick Demo Walkthrough	5 min	—
2	Domain & Custom Site	3 min	P1
3	Auth & Role-Based Portals	4 min	P1
4	Chrome Web Store Launch	3 min	P1
5	Checkout / Payment Links	3 min	P1
6	Newsletter System	3 min	P2
7	Careers Platform	2 min	P2
8	AI Agent (Text & Speech)	3 min	P3
9	Company Email & Collaboration	2 min	P3
10	Budget & Next Steps	2 min	—

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1. Domain & Custom Website (Ent-01)

What

Register `thingsthatmatter.co.za` (or `.com`) and point it to our GitHub Pages hosted site. Professional web presence instead of `github.io` URL.

Why

- Brand credibility for investors, merchants, and users
- Required for Chrome Web Store developer verification
- Professional email addresses (e.g., `hello@thingsthatmatter.co.za`)
- SEO benefits from custom domain

Recommended Approach

GitHub Pages + Custom Domain (already supported on free GitHub plan)

Steps:

1. Register domain at domains.co.za (~R80/year for `.co.za`)
2. Configure DNS records at registrar:
   - `A` records → GitHub's IPs: `185.199.108.153`, `185.199.109.153`, `185.199.110.153`, `185.199.111.153`
   - `CNAME` record: `www` → `thingsthatmatter.github.io`
3. Add `CNAME` file to repo root with domain name
4. Enable HTTPS in GitHub Pages settings (free, automatic via Let's Encrypt)

Cost

Item	Cost
`.co.za` domain	~R80/year
`.com` domain (optional)	~R200/year
GitHub Pages hosting	Free
HTTPS certificate	Free (Let's Encrypt)

Timeline

1 day — Register domain, configure DNS, deploy.

Decision Needed from Sobuhle

- Domain name preference: `.co.za`, `.com`, or both?
- Exact domain: `thingsthatmatter.co.za`? `thingsthatmatter.com`? Other?

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2. Auth & Role-Based Access Control (Ent-02, Ent-03, Ent-04)

What

Login system with different portals for different user types: CEO, CTO, Merchant, Investor, Consumer. Each role sees only what they should see.

Why

- Investors need a dashboard showing traction metrics, not admin tools
- Merchants need product management, not company financials
- CEO needs oversight of everything
- Security: principle of least privilege

Recommended Approach

Supabase Auth + Custom Claims RBAC

Supabase is already in our stack. Their RBAC system works as follows:

Architecture:

```
User signs in (Google/Email/Phone)
    ↓
```

```
Supabase Auth issues JWT
    ↓
Custom Access Token Hook injects user_role into JWT
    ↓
Frontend reads role from JWT → shows correct portal
    ↓
Backend RLS policies check role → enforce data access
```

Role Structure:

Role	Portal	Access
`ceo`	Executive Dashboard	Full oversight, financials, investor metrics
`cto`	Admin Panel	Full technical, user management, analytics
`merchant`	Merchant Portal	Product listings, sales data, catalog upload
`investor`	Investor Mode	Traction metrics, growth charts, read-only
`consumer`	Shopping View	Browse, wishlist, purchase

Implementation Details:

1. Create `app_role` enum in PostgreSQL: `('ceo', 'cto', 'merchant', 'investor', 'consumer')`
2. Create `user_roles` table linking `auth.users` to roles
3. Create `role_permissions` table for granular access
4. Deploy Custom Access Token Auth Hook (Supabase Edge Function) to inject `user_role` into every JWT
5. Frontend: decode JWT with `jwt-decode` package, route to correct portal
6. Backend: `authorize()` function in RLS policies checks `(auth.jwt() -> 'user_role')::text`

Auth Methods Supported (Free Tier):

- Email/password
- Magic link (passwordless)
- Google OAuth
- GitHub OAuth
- Phone (OTP/SMS)
- Apple Sign-In

Cost

Item	Cost
Supabase Free Tier	R0 (up to 50,000 MAU)
Supabase Pro (if needed)	~~$25/month~~ (R450/month)

Timeline

Sprint 7 (1 week) — Auth system + 2 portals (CEO + Consumer) **Sprint 8 (1 week)** — Merchant portal + Investor mode

Decision Needed from Sobuhle

- Which auth methods to enable first? (Recommend: Google + Email)

- Investor portal: what metrics should be visible?
- Should merchants self-register or be invited?

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3. Chrome Web Store Launch (Ent-05)

What

Publish the Lens extension (v0.3.0) to Chrome Web Store so users can install with one click and get automatic updates.

Why

- **Trust signal**: Google reviews the extension, users see "verified" badge
- **Auto-updates**: Bug fixes and features push automatically
- **Discovery**: Users can find us by searching "ethical shopping" in Chrome Web Store
- **Distribution**: One-click install instead of manual `.zip` sideloading
- **Analytics**: Store provides install counts, ratings, and user feedback

Recommended Approach

Chrome Web Store Developer Dashboard

Steps:

1. Pay one-time developer registration fee ($5 / ~R90)
2. Upload `extension-v0.3.0.zip` (already built and ready)
3. Fill store listing:
    - **Category**: Shopping
    - **Description**: Ethical shopping companion (already drafted)
    - **Screenshots**: 4 screenshots ready (1280×800px)
    - **Privacy tab**: Declare permissions and data usage
4. Submit for review (typically 1-3 business days)
5. Use **deferred publishing** to control exact launch date

Best Practices (from Google):

- MV3 required (we already use MV3 ✓)
- HTTPS for all network requests (we do ✓)
- Declare all permissions in privacy tab
- Sign In with Google integration (aligns with our auth strategy)
- Store listing should include a demo video (we're making one)

Cost

Item	Cost
Developer registration	$5 one-time (~R90)
Ongoing hosting	Free

Timeline

2 days — Submit and wait for review.

Decision Needed from Sobuhle

- Launch timing: Submit now or wait for auth to be integrated?
- Store listing: Any specific branding language for the description?

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4. Checkout / Payment Links (Ent-06)

What

Enable actual purchases through the extension's "Buy" buttons. When a user finds an ethical alternative, they can buy it directly.

Why

- Primary revenue model: commission on ethical purchases
- Reduces friction from "I found this product" to "I bought this product"
- Measurable conversion metrics for investors

Recommended Options

Platform	Best For	Commission	SA Support	Notes
PayFast	SA businesses	3.5% + R2 per transaction	✓ Native	SA-based, Rand payments, instant settlement
Stripe	Global scale	2.9% + R5.50 per transaction	✓ (since 2023)	Industry standard, global reach, excellent API
Lemon Squeezy	Digital products	5% + 50¢	✓	Merchant of record (handles VAT, tax)
Yoco	SA SMEs	2.6% - 3.18%	✓ Native	Very popular with SA small businesses

Recommended Approach

PayFast for MVP (South African market first), then add **Stripe** for global expansion.

Why PayFast first:

- South African merchants are our first customers
- Rand-denominated (no currency conversion fees)
- Instant settlement to SA bank accounts
- Easy payment link generation (no complex integration needed)
- Supports: card, EFT, SnapScan, Zapper, Masterpass

Integration:

1. Merchant signs up on PayFast
2. We generate affiliate/payment links per product
3. Extension's "Buy" button opens PayFast checkout
4. We track conversions via PayFast IPN (Instant Payment Notification)

Cost

Item	Cost
PayFast account	Free to set up
Per transaction	3.5% + R2
Stripe (later)	2.9% + R5.50 per transaction

Timeline

Sprint 8-9 — PayFast integration + conversion tracking

Decision Needed from Sobuhle

- Revenue model: Commission-based, subscription, or hybrid?
- Which payment providers do our target merchants already use?
- PayFast first, or go straight to Stripe?

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5. Newsletter System (Gr-01)

What

Email newsletter for user engagement, merchant updates, investor communications, and product announcements.

Why

- Direct communication channel (not dependent on social media algorithms)
- Investor updates (monthly traction reports)
- Merchant onboarding sequences
- User retention and re-engagement
- Build an email list as a company asset

Recommended Approach

Buttondown — API-first newsletter platform

Why Buttondown over Mailchimp/ConvertKit:

- **API-first**: Integrates with our existing TypeScript codebase
- **Markdown support**: Write newsletters in the same format as our docs
- **Paid subscriptions**: Can monetize premium content (keep 100% revenue)
- **Automations**: Onboarding drip sequences for merchants
- **GDPR-compliant analytics**: Open rates, click tracking, subscriber growth
- **RSS integration**: Auto-send when we publish blog posts
- **Surveys**: 200%+ higher response rates than standalone survey tools
- **Archives**: Public newsletter archive as web content (SEO)
- **Unlimited contributors**: Multiple team members with role-based permissions

Newsletter Segments:

Segment	Audience	Frequency	Content
Product Updates	All users	Bi-weekly	New features, tips
Merchant Brief	Merchants	Weekly	Sales data, new tools

Investor Report	Investors	Monthly	Metrics, milestones
Team Internal	Team	Weekly	Sprint updates, decisions

Cost

Plan	Cost	Subscribers
Free	R0	Up to 100
Basic	$9/month (~R165/month)	Unlimited
Professional	$29/month (~R530/month)	+ Automations, surveys

Timeline

1 week — Set up account, design templates, first newsletter

Decision Needed from Sobuhle

- Start free tier or go Basic immediately?
- Who writes content? CTO, CEO, or both?
- First newsletter topic?

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6. Careers Platform (Gr-02)

What

A `/careers` page on the website listing open positions, company culture, and application process.

Why

- Professional signal to investors ("this company is hiring")
- Attract talent as we scale
- Transparent company culture messaging
- Required for enterprise credibility

Recommended Approach

Static careers page on GitHub Pages (MVP) → **Lever or Ashby** later

MVP Approach (Free):

- Add `/careers` route to our GitHub Pages site
- List positions as Markdown files
- Application via Google Forms or email
- Cost: R0

Scale Approach (Later):

Platform	Cost	Best For
Lever	$$$	Enterprise hiring
Ashby	$$	Startup-friendly ATS

Workable	$149/month	Mid-size teams
Google Forms	Free	MVP/early stage

Cost

Item	Cost
MVP (static page + Google Forms)	Free
Ashby (when scaling)	~$300/month

Timeline

2 hours for MVP static page

Decision Needed from Sobuhle

- What positions to list? (Even aspirational ones signal growth)
- Application process: email or form?
- Company culture messaging — what values to highlight?

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7. AI Agent — Text & Speech (Sys-01)

What

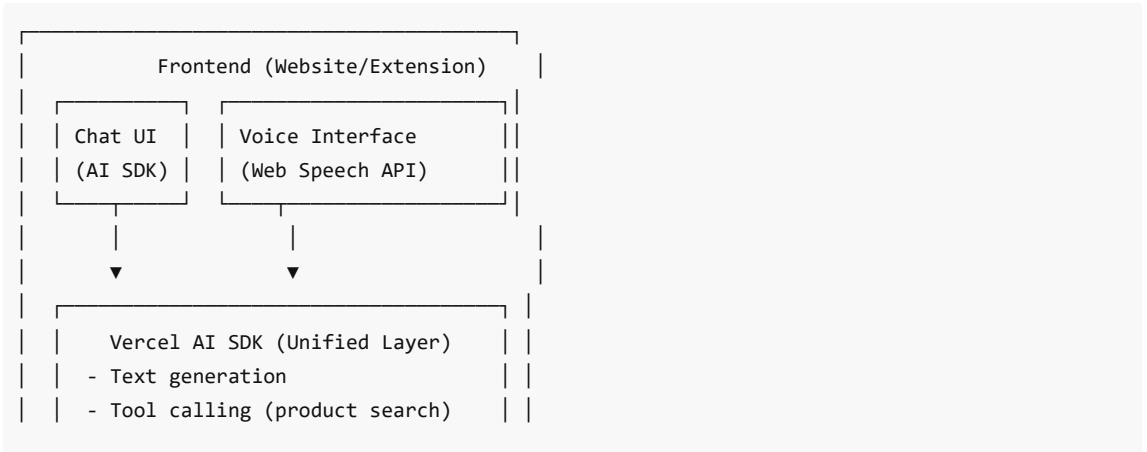
An AI-powered assistant on the website and/or extension that helps users find ethical products, onboard merchants, and answer questions — via text chat and voice.

Why

- Company tagline: "*For humans today, AI agents tomorrow*"
- Reduces support burden (FAQ, onboarding)
- Differentiator: most ethical shopping tools don't have AI assistance
- Voice interface increases accessibility
- Positions us as AI-forward for investors

Recommended Architecture

Three-layer approach:



```
│  │  - Streaming responses                │ │
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│                 ▼                          │
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│  │   LLM Provider (swappable)           │  │
│  │  - OpenAI GPT-4o-mini (cheap)        │  │
│  │  - Anthropic Claude (quality)        │  │
│  │  - Groq (fast, free tier)            │  │
│  └──────────────────────────────────────┘  │
└────────────────────────────────────────────┘
```

Layer 1: Vercel AI SDK (Text Chat)

Why Vercel AI SDK:

- TypeScript-native (matches our stack)
- Provider-agnostic: swap between OpenAI, Anthropic, Groq without code changes
- Built-in streaming (real-time chat responses)
- Tool calling: agent can search our product catalog, look up merchant info
- UI hooks: `useChat()` gives you a full chat interface in 10 lines of React
- MCP (Model Context Protocol) support: connects to external data sources

Code Example:

```
import { generateText } from "ai";
import { openai } from "@ai-sdk/openai";

const { text } = await generateText({
  model: openai("gpt-4o-mini"),
  system: "You are .things that Matter's ethical shopping assistant.",
  prompt: userMessage,
  tools: {
    searchProducts: { /* search our catalog */ },
    findAlternative: { /* find ethical alternatives */ },
  }
});
```

Cost:

- GPT-4o-mini: ~$0.15 per 1M input tokens (extremely cheap)
- Groq (Llama 3): Free tier available
- Claude Haiku: ~$0.25 per 1M input tokens

Layer 2: Web Speech API (Voice — Free, Browser-Native)

Speech-to-Text (STT):

- Built into Chrome, Edge, Safari — no API costs
- `SpeechRecognition` interface: user speaks → text
- Supports continuous recognition and interim results
- Works offline with Chrome's on-device recognition (experimental)

Text-to-Speech (TTS):

- `SpeechSynthesis` interface: text → spoken voice
- Multiple voice options per language
- Free, no API calls, runs locally in browser

Code Example:

```
// Speech to Text
const recognition = new SpeechRecognition();
recognition.lang = 'en-ZA';  // South African English
recognition.onresult = (event) => {
  const userSaid = event.results[0][0].transcript;
  // Send to AI SDK for processing
};
recognition.start();

// Text to Speech
const utterance = new SpeechSynthesisUtterance(aiResponse);
speechSynthesis.speak(utterance);
```

Browser Support: Chrome ✓, Edge ✓, Safari ✓ (covers 90%+ of users)

Layer 3: Premium Voice (Future — ElevenLabs)

For a branded, human-like voice (future phase):

Service	Cost	Quality
ElevenLabs	$5/month (30 min)	Near-human voice cloning
OpenAI TTS	$15 per 1M chars	High quality, 6 voices
Web Speech API	Free	Robotic but functional

Cost

Component	Cost
Vercel AI SDK	Free (open source)
Web Speech API	Free (browser native)
GPT-4o-mini (1000 chats/day)	~~$5/month~~ (R90/month)
Groq free tier	R0 (rate limited)
ElevenLabs (future)	$5/month (~R90/month)

Timeline

Sprint 9-10 — Text chat MVP with product search tools **Sprint 11** — Voice interface with Web Speech API **Sprint 12** — Premium voice + advanced agent capabilities

Decision Needed from Sobuhle

- Start with text-only or include voice from day one?

- LLM provider preference: cheapest (Groq) vs. best quality (Claude)?
- Agent personality: formal, friendly, Gen-Z, or professional?
- Should the agent have a name? (e.g., "Matter", "Thandi", etc.)

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8. Company Email & Collaboration System (Sys-02)

What

Professional email addresses (`matome@thingsthatmatter.co.za` , `sobuhle@thingsthatmatter.co.za`) and internal collaboration tools (calendar, contacts, file sharing) — self-hosted for IP security.

Why

- Professional communication (not `@gmail.com`)
- IP security: emails about business strategy, investor conversations, and code discussions stay on our servers
- Required for: Chrome Web Store verification, merchant trust, investor confidence
- Calendar/contact sync for team coordination

Options Comparison

Solution	Type	Cost	Pros	Cons
Google Workspace	Cloud (Google)	$6/user/month (~R110/user)	Easy, reliable, Google ecosystem	Google reads your email, not self-hosted
Zoho Mail	Cloud	Free (5 users), $1/user/month	Cheap, good UI	Limited free tier
Stalwart Mail	Self-hosted	Free (open source)	Full control, IP secure, Rust-built	Requires server management
Proton Mail Business	Cloud (encrypted)	$8/user/month (~R145/user)	End-to-end encrypted, Swiss privacy	Expensive, limited integrations

Recommended Approach

Phase 1 (Now): Zoho Mail Free Tier

- 5 users, 5GB each, custom domain
- Professional email immediately
- Calendar and contacts included
- Cost: R0

Phase 2 (Growth): Stalwart Mail Server (Self-Hosted)

Why Stalwart:

- **Written in Rust** (memory safe, matches our tech DNA)
- **All-in-one**: SMTP, IMAP, JMAP, anti-spam, calendars, contacts, file sharing — single binary
- **Encryption at rest**: S/MIME or OpenPGP for all stored emails
- **Anti-spam**: DMARC, DKIM, SPF, ARC, greylisting, DNS blocklists, phishing protection
- **Collaboration**: CalDAV (calendar), CardDAV (contacts), WebDAV (file sharing)
- **Scalable**: Can handle millions of mailboxes with FoundationDB backend
- **Web interface**: Built-in admin panel and self-service portal

- **Security audited**: Commercial-grade security

Stalwart Deployment:

1. Run on a VPS (Hetzner R85/month for 4GB RAM)
2. Configure MX records to point to our server
3. Set up DKIM/DMARC/SPF for deliverability
4. Enable TLS with ACME (automatic certificates)
5. Web admin panel for user management

Cost

Phase	Item	Cost
Phase 1	Zoho Mail Free	R0 (up to 5 users)
Phase 2	Hetzner VPS	~R85/month
Phase 2	Stalwart Community	Free (open source)
Phase 2	Stalwart Enterprise	Contact for pricing

Timeline

Phase 1: 2 hours (Zoho setup + DNS) **Phase 2:** Sprint 12+ (Stalwart deployment when team grows)

Decision Needed from Sobuhle

- Start with Zoho free tier or go straight to self-hosted?
- How many email accounts needed now? (CEO, CTO, info@, hello@?)
- Priority: speed (Zoho) or IP security (Stalwart)?

Budget Summary

Immediate Costs (This Month)

Item	Cost	Priority
Domain registration (.co.za)	R80/year	P1
Chrome Web Store registration	R90 one-time	P1
Zoho Mail (free tier)	R0	P1
Total immediate	**~R170**	

Monthly Operating Costs (After Launch)

Item	Monthly Cost	When
Supabase (free tier)	R0	Sprint 7
Newsletter (free tier)	R0	Sprint 7
AI chat (GPT-4o-mini)	~R90	Sprint 9

VPS for email (Stalwart)	~R85	Sprint 12
Newsletter (Basic)	~R165	When > 100 subs
Supabase Pro	~R450	When > 50K users
Total monthly (projected)	**R0 → R790/month**	

Current R3,500 Budget Allocation

Item	Amount
Cursor AI subscription	R665
Domain + Chrome Store registration	R170
Development tools (remaining)	R2,665
Total	**R3,500**

Recommended Sprint Roadmap

Sprint	Focus	Deliverables
Sprint 7 (Week 1)	Foundation	Domain live, Zoho email, Auth system, Google OAuth
Sprint 8 (Week 2)	Commerce	Chrome Web Store live, PayFast integration, Merchant portal
Sprint 9 (Week 3)	Intelligence	AI text assistant, Newsletter launch, Investor mode
Sprint 10 (Week 4)	Growth	Voice interface, Careers page, Analytics dashboard

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Questions for Sobuhle

1. **Domain**: `.co.za` , `.com` , or both?
2. **Auth**: Google + Email first, or include Phone/SMS?
3. **Chrome Store**: Submit now or wait for auth integration?
4. **Payments**: PayFast (SA-first) or Stripe (global-first)?
5. **Revenue model**: Commission, subscription, or hybrid?
6. **Newsletter**: Who writes the first edition?
7. **AI agent**: Name and personality?
8. **Email**: Zoho free tier now, or self-host from day one?
9. **Investor metrics**: What numbers matter most to show?
10. **Hiring**: Any positions to list on careers page?

Prepared for the Monday 17 Feb meeting. All research backed by current documentation from GitHub, Supabase, Google, PayFast, Buttondown, Vercel AI SDK, and Stalwart Labs.